BLOCK BLAST

Project Workflow: Week 1

PHASE: PROJECT PLANNING & CONCEPT FINALIZATION

1. Objective

The primary goal of Week 1 was to establish a robust project foundation. This included:

- Defining the game scope, mechanical rules, and structural boundaries.
- Selecting an optimal technology stack for grid manipulation, event handling, and 2D rendering performance.
- Conceptualizing the UI/UX design framework to deliver clean visual feedback and intuitive drag-and-drop mechanics.

2. Detailed Task Breakdown

- **Feature & Mechanic Identification:** Conducted thorough research on puzzle game standards to finalize the core gameplay mechanics, establishing a defined 8×8 grid matrix, custom polyomino block shape generation patterns, full row/column clear checking algorithms, and strict game-over boundary criteria.
- **Technical Stack Selection:** Selected Java Swing and AWT frameworks for their stability, native UI component control, flexible paint overrides via *Graphics2D*, and reliable mouse interaction handling.
- **UI/UX Design:** Created initial layout wireframes comprising a central game board layout, a bottom 3-slot dock for incoming block choices, and an persistent top score tracker panel. Formulated a cohesive dark visual scheme (Background: `#18181B`) accented with vibrant, high-contrast block colors to maximize readability.
- **Architectural Setup:** Formulated a modular code blueprint that effectively separates core engine state tracking (grid state management, collision verification, line clearing) from display management and event listening hooks.

3. Deliverables

- **Project Scope Document:** A formal architectural specification highlighting gameplay loop mechanics, score multipliers, and engine constraints.
- **UI Wireframe Sketches:** Draft layouts displaying component alignments for the primary matrix, block docks, and game-over states.
- **Color Palette Definition:** A production-ready AWT Color mapping dictionary detailing background colors, grid borders, and individual block hex keys.

- **Project Timeline:** A structured, 10-week end-to-end development milestone chart ranging from initialization to final playtesting and binary compilation.

4. Tools & Resources

DEVELOPMENT ENVIRONMENT

- Java JDK 11+
- IntelliJ IDEA / VS Code
- Git (Version Control)

REFERENCE MATERIALS

- Java Swing & AWT `Graphics2D` APIs
- 2D Matrix Manipulation & Overlap Algorithms